

Explain Any Dataset as per your choice and its exploration using R

Dataset Name: Food Delivery Dataset (food_delivery.csv)

Objective

The objective of this assignment is to analyze the food delivery dataset using R programming. This helps in understanding order patterns, delivery time, customer ratings, and performance of restaurants.

Step 1: Set Working Directory

Set the location where the dataset file is stored.

```
setwd("C:/Users/tarla/OneDrive/Desktop/5029")
```

Step 2: Load Dataset

Load the dataset into R.

```
data <- read.csv("food_delivery.csv")
```

Path:

```
data <- read.csv("C:/Users/tarla/OneDrive/Desktop/5029/food_delivery.csv")
```

Step 3: Explore the Dataset

View the structure and summary of the dataset.

```
head(data)
```

```
str(data)
```

```
summary(data)
```

Step 4: Disable Scientific Notation

```
options(scipen = 999)
```

Step 5: Top 10 Restaurants by Orders

```
dev.new()
```

```
top10_rest <- sort(table(data$Restaurant), decreasing = TRUE)[1:10]
```

```
par(mar = c(10,4,4,2))
```

```
barplot(top10_rest,  
        col = "orange",  
        las = 2,  
        main = "Top 10 Restaurants by Orders",  
        ylab = "Number of Orders")
```

Output:

Bar chart showing top restaurants with highest number of orders.



Step 6: Delivery Time Distribution

```
dev.new()
```

```
hist(data$Delivery_Time,  
      col = "skyblue",  
      main = "Delivery Time Distribution",  
      xlab = "Time (minutes)")
```

Output:

Histogram showing how delivery time is distributed.



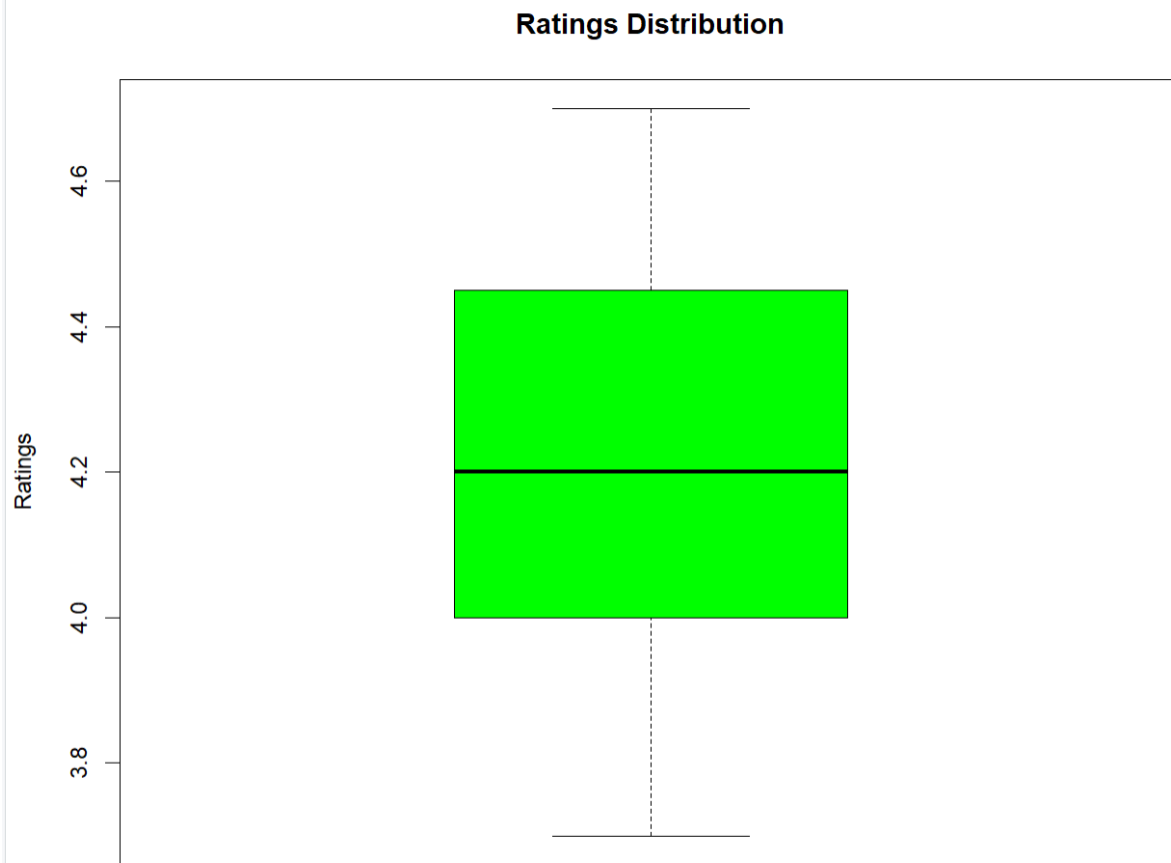
Step 7: Ratings Analysis

```
dev.new()
```

```
boxplot(data$Rating,  
        col = "green",  
        main = "Ratings Distribution",  
        ylab = "Ratings")
```

Output:

Boxplot showing spread of customer ratings.



Step 8: Pie Chart: Distribution of Orders Across Different Cities

```
library(plotrix)
```

```
data <- read.csv("C:/Users/tarla/OneDrive/Desktop/5029/food_delivery.csv")
```

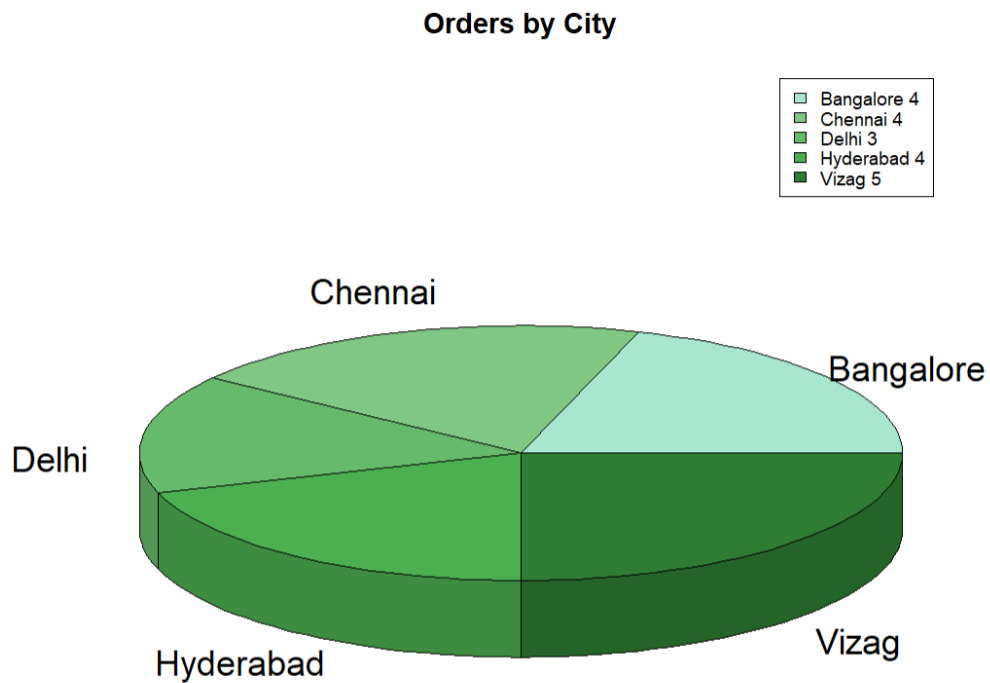
```
city_orders <- table(data$City)
```

```
colors <- c("#A8E6CF", "#81C784", "#66BB6A", "#4CAF50", "#2E7D32")
```

```
pie3D(city_orders,
      labels = names(city_orders),
      col = colors,
      main = "Orders by City")

legend("topright",
      legend = paste(names(city_orders), city_orders),
      fill = colors,
      cex = 0.8)
```

Output:



Step 9: Scatter Plot Showing Relationship Between Delivery Time and Ratings

```
data <- read.csv("C:/Users/tarla/OneDrive/Desktop/5029/food_delivery.csv")
```

```
plot(data$Delivery_Time, data$Rating,  
     main = "Delivery Time vs Rating",  
     xlab = "Delivery Time (minutes)",  
     ylab = "Rating",  
     pch = 19,  
     col = "#4CAF50")  
  
grid()
```

Output:



Step 10: Line Graph Showing Number of Orders Across Cities

```
data <- read.csv("C:/Users/tarla/OneDrive/Desktop/5029/food_delivery.csv")
```

```
city_orders <- table(data$City)
```

```
plot(city_orders,
```

```
  type = "b",
```

```
  main = "Orders by City",
```

```
  xlab = "City",
```

```
  ylab = "Number of Orders",
```

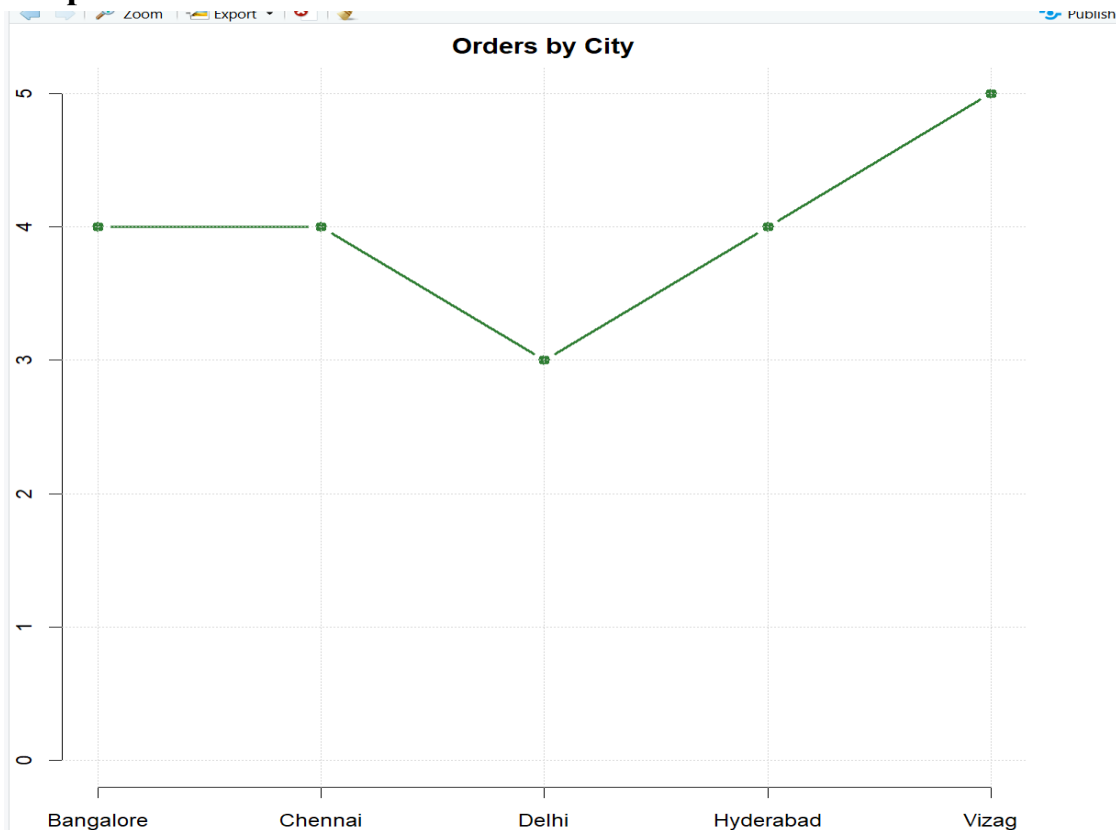
```
  col = "#2E7D32",
```

```
  lwd = 2,
```

```
  pch = 19)
```

```
grid()
```

Output:



Conclusion

From the analysis:

- Identified top restaurants based on number of orders
- Studied delivery time distribution
- Analyzed customer ratings
- Compared performance across cities
- Observed variation using boxplots

R programming provides powerful tools for data visualization and analysis, making it easier to understand real-world datasets.